

Statistical Report

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Statistical Report on Fertility Data in Singapore

Abstract

Introduction

Can time series models trained on data from 1960–2012 accurately forecast Singapore’s continuing fertility decline over 2013–2024?

TO INCLUDE WHEN INTRODUCTION IS DONE

Include necessary abbreviations when writing it up

4. Justify SARIMA period 12 in Introduction (page 3) The periodogram doesn’t obviously point to period 12, so you need literature to justify it. Add a paragraph in the Introduction discussing the Chinese zodiac Dragon year effect on Singapore births. This is what makes the SARIMA(p,d,q)(P,D,Q)[12] models substantively motivated rather than just data-driven.

Methods

The modelling workflow was designed to answer our forecasting needs. The TFR and TLB series were extracted from the dataset, reshaped into annual time series from 1960–2024 and split into a training period from 1960–2012 and a testing period from 2013–2024.

Time plots and the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests of the two series were used to check for trends, changes in variance and the transformations required before modelling. Both series showed evidence of non-stationarity and since ARIMA models require the time series to be stationary, transformation of the series was necessary prior to modelling.

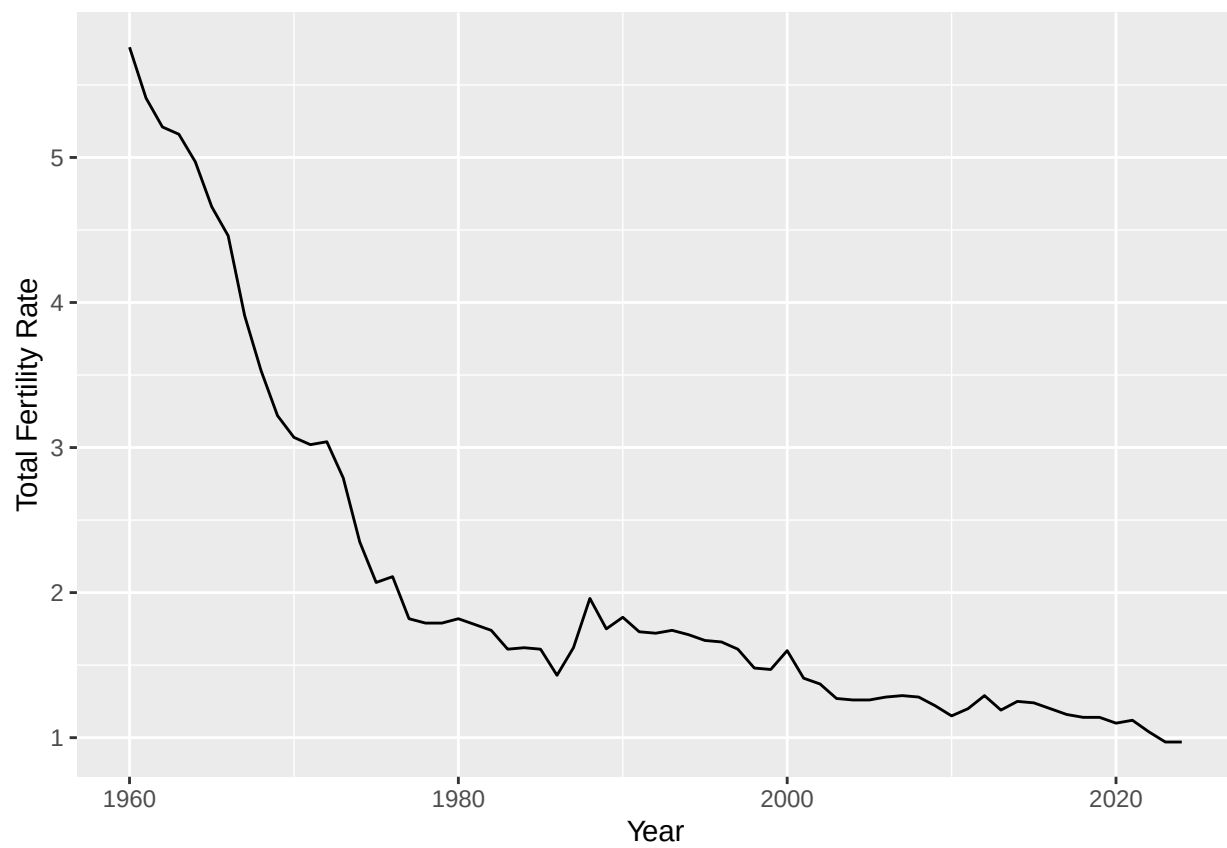
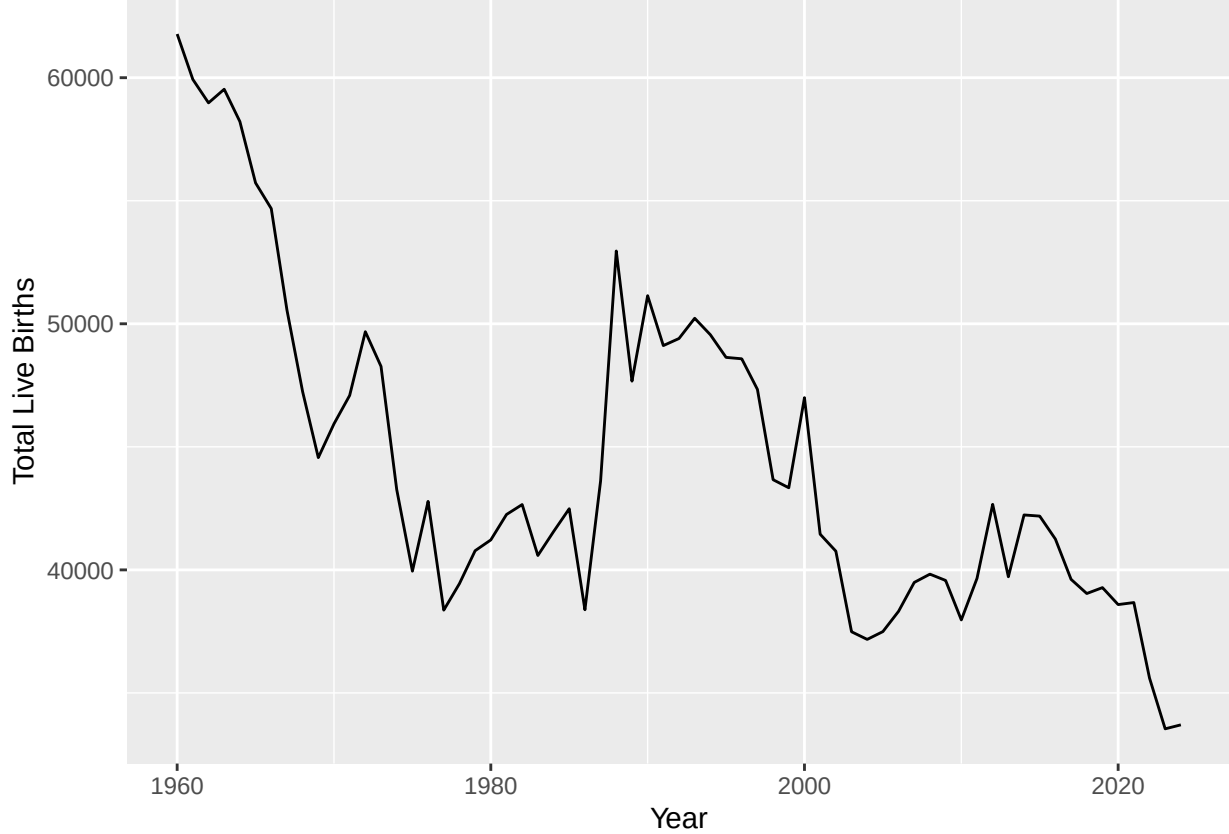


Figure 1: Annual Total Fertility Rate in Singapore



The log transformation was used to stabilise variance and make the model proportional rather than absolute changes. Differencing was used to remove the trends shown inside the data.

Both series were then transformed using either just the log or the log with first or second differencing. These transformed series were then tested by the KPSS test to ensure an improvement of the stationarity of the series.

Two transformations showed potential as being stationary (first-differenced and logged, and second differenced logged). With the KPSS test a small p-value shows evidence against stationarity. For TFR, the KPSS test gave that $p = 0.0406$ for the first differenced logged series and $p = 0.1$ for the second differenced logged series. For the logged TLB with both first and second differencing gave the same p-values of $p = 0.1$. So for TLB stationarity was not rejected for either transformation.

Despite the first difference of the TFR data having a KPSS test result below the significance threshold, we would prefer there to be less differencing as we do not want to over-difference the data and introduce noise in a relatively short data-set. We kept the second differenced models as comparison as well so we could see another set of models in the results.

Because the transformed series and baseline residual diagnostics showed dependence around lags 11–13, higher-lag ARIMA models were also considered. These models were included to capture longer-lag autoregressive dependence that was not represented by the low-order baseline models.

SARIMA-style models with period 12 were also considered. Although the data are annual, the period-12 structure was motivated by both the observed lag-12 dependence and the literature discussed in the Introduction on Chinese zodiac-related birth timing effects in Singapore. The SARIMA models were therefore included to test the 12 year dependence could be captured better by the seasonal model than the unrestricted high order ARIMA models. For our candidate models, the ACF and PACF of the transformed series, were inspected to guide our the first viable period-12 specification.

Models were only considered viable if their residuals were approximately white noise. This again was assessed with ACF and PACF and residual plots. Among viable models, the final model selection was based on AIC

and forecast accuracy over the 2013-2024 period. Forecast accuracy was assessed with RMSE and MAE, with formulas provided in the appendix.

- So informed decision to try ARMIA (12,x, y) and SARIMA models potentially

Series	Models
TFR	SARIMA(0,1,1)(1,0,0)[12], SARIMA(1,1,1)(1,0,0)[12], SARIMA(1,2,1)(1,0,0)[12]
TLB	SARIMA(0,1,1)(1,0,0)[12], SARIMA(1,1,1)(1,0,0)[12], SARIMA(0,2,1)(1,0,0)[12]

Results

Your professor expects the Results section to show:

ACF/PACF plots of the detrended and deseasonalised series
How the first viable model was identified from those plots
How you varied parameters to get additional viable models
How each viable model fits the testing data

Discussion

Conclusion

References

Appendices if applicable.